Mechanistic Estimation of Neural Networks: Deep Third-Cumulant Content

Round 1 Proposal

Q1. A representation that avoids per-old-source dense Hadamard contractions

Derived / proposal: reverse-mode adjoint through the analytic mean–covariance propagation.

The key observation is that the ReLU closure at layer l only needs the slices

$$D21_{ic}^{(l)} = \kappa_3(z^{(l)})_{iic}, \qquad D3_i^{(l)} = \kappa_3(z^{(l)})_{iii}$$

of the pre-activation third cumulant. The final mean $\mu^{(L)} = E[h^{(L)}]$ is a linear functional of these slices, because the third cumulant enters the covariance update linearly. Therefore, instead of forward-transporting every source, we can carry the **adjoint sensitivities** of the final loss with respect to these slices.

Object carried

Forward pass: only

$$\mu^{(l)} = E[h^{(l)}] \in \mathbb{R}^n, \qquad C^{(l)} = \mathrm{Cov}(h^{(l)}) \in \mathbb{R}^{n\times n}.$$

No explicit κ_3 sources are propagated.

Backward pass: for each layer l carry

$$S^{(l)} = \frac{\partial \mathcal{L}}{\partial D21^{(l)}} \in \mathbb{R}^{n\times n}, \qquad s^{(l)} = \frac{\partial \mathcal{L}}{\partial D3^{(l)}} \in \mathbb{R}^n,$$

where $\mathcal{L} = \|\mu^{(L)} - \mathrm{target}\|^2$. These are exactly the adjoints of the slices needed by the ReLU closure.

Per-layer update

Forward:

$$z^{(l+1)} = W^{(l+1)}h^{(l)}, \qquad \mu^{(l+1)} = E[\mathrm{ReLU}(z^{(l+1)})], \qquad C^{(l+1)} = \mathrm{Cov}(\mathrm{ReLU}(z^{(l+1)})).$$

The covariance update uses a Gaussian closure plus a correction linear in $D21^{(l+1)}, D3^{(l+1)}$.

Backward: the adjoint of the transport

$$\kappa_3(z^{(l+1)})_{iic} = \sum_{a,b,c} W_{i,a} W_{i,b} W_{c,c} \kappa_3(h^{(l)})_{abc}$$

maps $(S^{(l+1)}, s^{(l+1)})$ to a sensitivity with respect to $\kappa_3(h^{(l)})$. Because the ReLU is elementwise, the adjoint of the ReLU closure then maps this to a sensitivity with respect to $D21^{(l)}, D3^{(l)}$. The whole backward step per layer is one dense contraction of an $n \times n$ matrix with W (cost n^3 FLOPs) plus $O(n^2)$ elementwise work.

Cost in units. n^3 matmul = 1 unit. One backward pass per layer costs ≈ 1 unit (plus small elementwise). For $L = 16$, total ≈ 16 units. This is independent of the number of old sources. The leaders' bill (133–165 units) is roughly V29's young tier (153 units) plus such a 16-unit backward pass. **Conjecture:** this is what they do.

What it loses

It is exact to the same Hermite closure order as V29: it computes the linear effect of the true κ_3 on the final mean without ever forming the full tensor. It does not approximate the slices; it avoids transporting them forward. The only approximation is the same memoryless 4th-cumulant regeneration already used in V29. It loses the ability to output per-layer κ_3 slices, but those are not needed for the final mean.

First falsifying experiment

Implement the adjoint on a small network ($n = 256$, $L = 8$, one fixed seed). Compare the final 256 means against the exact V29 sum-of-sources result. If the MSE difference is $> 10^{-8}$ (scaled), the adjoint is missing higher-order terms or the ReLU-closure adjoint is wrong. Runtime < 30 min on CPU. **Measure.**

Q2. Three ranked hypotheses for the leaders' "zero marginal cost" old sources

1. Adjoint sensitivity method (most likely)

What they carry: mean + covariance forward; sensitivity matrices $S^{(l)}$ backward.

Cost signature: cost per layer ≈ 1 unit, independent of the number of old sources. Total cost should scale as $O(Ln^3)$, not $O(Ln^2 r \cdot \#\text{sources})$.

Distinguishing observable: On the public leaderboard, if you increase depth L, their cost should grow linearly by ~ 1 unit per layer. If you increase the number of old sources (e.g., by changing the birth schedule), their cost should not change.

Local experiment: Implement the adjoint on one network and compare its FLOPs and MSE to V29. If it matches $2.1e-8$ at ~ 150 units, this is it. **Measure.**

2. Adaptive Tucker decomposition of the full third cumulant

What they carry: a single Tucker core $T^{(l)}$ of rank r_l per layer, with r_l decreasing from 384 at layer 4 to 100 at layer 14.

Cost signature: per-layer cost nr_l^3 (with factor 2 for float64). At $r = 100$, $nr^3 \approx 1$ unit. Total ~ 16 units.

What it loses: section 3 says cliffs below $r \sim 150$. But if the leaders use a different basis or a nonlinear closure that avoids cliffs, they could use $r = 100$.

Distinguishing observable: Their error should degrade smoothly with depth if they use adaptive rank. You can measure the rank of the true third cumulant tensor (not legs) at each layer. If the true tensor has effective rank ~ 100 at layer 14, this is plausible. **Measure the Tucker rank of the full κ_3 tensor on one network.**

3. Randomized sketch of the Hadamard contractions

What they carry: for each source, a random projection of $A \circ P$ onto $k \ll n$ columns, with k chosen to reach 2% RMS error.

Cost signature: $O(knr)$ per source per layer. If $k \approx 100$, cost ≈ 0.1 unit per source.

Why less likely: section 3 shows the Hadamard spectrum is flat; to get 2% error you need $k \sim n$. So no saving.

Distinguishing observable: Their error should show stochastic variance across seeds. Public leaderboard MSE would fluctuate more than V29. **Measure seed-to-seed variance.**

Q3. If nothing is credible, the next measurement

If the adjoint and adaptive-Tucker hypotheses fail, the single most informative measurement is:

Measure the exact sensitivity of the final mean to the third cumulant slices at each layer, for a single network.

Concretely: take one trained network. For each layer l, perturb $D21^{(l)}$ and $D3^{(l)}$ by a small random matrix/vector, run the exact V29 forward propagation, and measure the change in the final 1024 means. This gives the empirical sensitivity tensor $S^{(l)}$. Then compute its rank (or its effective rank via singular values). If $S^{(l)}$ is low-rank (e.g., rank ≤ 50), then a factored backward pass is cheap and the adjoint method is viable. If $S^{(l)}$ is full-rank, then the leaders must be using a different trick.

Cost: 16 layers $\times$ 10 perturbations $\times$ one forward pass (260 units) = 41,600 units = $41{,}600 \times 2^{31}$ FLOPs $\approx 8.9e13$ FLOPs. On a CPU at $1e11$ FLOP/s, this is $\sim$ 15 minutes. **Measure.** This directly tests whether the deep third cumulant's effect on the final mean is compressible in the adjoint representation, which is the only way to get it at near-zero marginal cost per old source-layer.

Round 2: Deep Third-Cumulant Content in Random ReLU MLPs

Q1. Aggregate Response State: Fitting Procedure

Object. Carry $T = \sum_{a=1}^{k} \mathrm{sym3}(G_a \otimes v_a)$, $G_a \in \mathbb{R}^{n\times n}$ symmetric, $v_a \in \mathbb{R}^n$. Slices:

$$D21_{ic} = \sum_a \big((G_a)_{ii}(v_a)_c + 2(G_a)_{ic}(v_a)_i\big), \quad D3_i = 3\sum_a (G_a)_{ii}(v_a)_i.$$

Transport under $B = W\,\mathrm{diag}(w1)$: $G_a \leftarrow BG_aB^T$, $v_a \leftarrow Bv_a$. Cost per component per layer: two n^3 matmuls = 2 units, plus $n^2 = 0.001$ unit. Total $2k$ units/layer.

Fitting at age gate. At the layer where a source becomes old (age 4), we have exact $D21, D3$ from the source sum. Minimize relative error:

$$\min_{\{G_a, v_a\}} \|D21 - D21_{\text{approx}}\|_F^2/\|D21\|_F^2 + \|D3 - D3_{\text{approx}}\|_2^2/\|D3\|_2^2.$$

Use alternating least squares. Initialize v_a from top singular vectors of $D21$. For fixed v_a, the problem is linear in G_a. Solve by gradient descent (500 iterations). Cost per iteration $O(kn^2) \approx 10$ MFLOP for $k = 10$; total fitting < 1 GFLOP. **Measure-this.**

Expected k. From M3, the exact contracted query U has rank ~ 230 for 2% error. Our form is less expressive. Conjecture: $k \approx$ 20–50 needed for 2% on $D21$. **Measure-this:** fit $k = 1, 5, 10, 20, 50$ and record relative error.

Falsifier. Take one network, layer 6 (age gate for source born at 2). Extract exact $D21, D3$. Fit k components. If relative error $> 2\%$ for $k \leq 10$, the idea fails. If $< 2\%$ for $k \leq 5$, proceed to transport and check final MSE. Runtime < 1 hour.

Q2. Non-source formulations

(a) Reverse-mode adjoint. *Derived/closed:* adjoint sensitivity gives no cost drop because the forward covariance update still needs $D21, D3$ at every layer. Reject.

(b) Characteristic function in random-feature basis. Carry $\phi(t) = E[e^{it\cdot z}]$ projected on a basis of size m. ReLU update requires a high-dimensional convolution; no known $O(n^2)$ update. Reject.

(c) Quenched random tensor with known covariance. By T1, $E[\kappa_3|\text{even}] = 0$. The final mean is linear in the realized odd part. Carrying only covariance loses the sign and cannot compute the mean. Reject.

(d) Tensor-network contraction. Contract birth index j first: equivalent to forming dense legs. No saving. Reject.

(e) TAP/cavity. Standard TAP gives $\kappa_3 = 0$. Quenched odd structure requires higher-order cavity equations, likely $O(n^3)$ per layer. Reject.

New proposal: hierarchical Tucker of the full third cumulant. Carry κ_3 as a hierarchical Tucker tensor with ranks r_l per layer. Update under B: contract each core with B. Cost per layer $O(nr_l^2 + r_l^3)$. If $r_l \sim 100$, cost ~ 1 unit/layer. Loss: same as Tucker confinement (14–31x worse in closed list), but here ranks are adaptive and not confined to a shared basis. **Measure-this:** compute hierarchical Tucker ranks of exact κ_3 on one network.

Q3. Leaders' likely method

Conjecture. Leaders use a hierarchical Tucker or tensor-train representation of the full third cumulant, updated in compressed form without source decomposition. Cost $O(nr^2)$ per layer with $r \sim 100$, giving ~ 1 unit/layer. Total ~ 16 units for deep content. Combined with an optimized young tier, total 133–165 units.

Observable. On the public leaderboard, their cost should scale as $O(L)$ with a small constant independent of the number of sources. Their error should degrade smoothly with rank r. **Measure-this:** implement a tensor-train propagation on one network and compare MSE and FLOPs.

Strongest argument if nothing remains

If hierarchical Tucker fails, the strongest argument is: M2 shows every layer and every birth matters. Any method that carries deep content must store it (cost > 0) or compute it on the fly (cost > 0). The leaders' cost equals the young tier alone, so they must have found a representation where deep content is computed as a byproduct of the young tier. No such representation is currently known.